

Fiat Currency and Cryptocurrency: Two Sides of the Same Coin?

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Abstract

This study documents foreign exchange (FX) volatility as a positively priced cross-asset risk factor in the cross-section of cryptocurrency excess returns, implying that investors demand a premium for holding cryptocurrencies whose payoffs co-move with FX volatility under a standard risk-adjustment framework. To underscore the practical implications of this pricing power, we show that cross-asset time-series trading strategies conditioned on historical FX volatility are profitable. These findings align well with the currency competition theoretical framework and offer important insights for both academics and practitioners.

JEL Classification: G12, G15, F31

Keywords: cryptocurrency returns; FX volatility; asset pricing; trading strategy.

1 Introduction

Digital currencies, commonly known as cryptocurrencies, have recently garnered significant academic attention, spurring research into the cross-section of their expected returns (e.g., [Zhang et al., 2021](#); [Liu et al., 2022](#); [Borri and Shakhnov, 2022](#); [Fieberg et al., 2023](#); [Lee and Wang, 2024](#), among others). A common thread across these studies is their reliance on crypto-specific risk factors to explain the cross-sectional variation in cryptocurrency risk premia, consistent with standard asset-pricing paradigms in the [Fama and French \(1993\)](#) and [Carhart \(1997\)](#) models adapted to digital assets. A critical review of this small but growing literature reveals that, to our knowledge, no prior study has incorporated cross-market data—such as traditional risk factors from equities, bonds, or currencies—to price cryptocurrency premia, and the present study aims to address this gap.

Our primary research question investigates whether FX volatility risk is priced in cryptocurrency returns within a standard asset-pricing framework. We assess the cross-sectional implications of this question using a four-factor model based on FX volatility risk—serving as a proxy for global macroeconomic uncertainty ([Menkhoff et al., 2012a](#))—augmented by crypto market, size, and momentum factors, established in [Liu et al. \(2022\)](#) as robust proxies for systematic risk in cryptocurrency markets. The FX volatility risk factor, constructed from 10 major developed-market currencies, is tested against weekly equally weighted excess returns of quintile cryptocurrency portfolios spanning June 2, 2014, to October 28, 2024.

Our central finding is that FX volatility risk is priced in the cross-section of cryptocurrency returns, indicating that it captures information relevant for pricing not only fiat currencies ([Menkhoff et al., 2012a](#)) but also digital assets, implying common systematic factors underpin cross-sectional pricing across crypto and fiat currency markets. We find that cryptocurrency returns exhibit positive, significant FX volatility risk premia, suggesting that cryptocurrencies with higher exposure to FX volatility have significantly higher returns than cryptocurrencies with lower exposure to this factor. Within a standard risk–return framework, our evidence of a significantly positive risk premium on FX volatility risk (macro uncertainty proxy) implies that crypto investors are compensated for bearing exposure to this global FX-related risk.

We validate this novel evidence through comprehensive robustness tests across multiple specifications. The FX volatility factor retains significant pricing ability when test assets shift from equal- to value-weighting, FX volatility construction incorporates emerging currencies, analysis restricts to turmoil/tranquility subsamples, and individual cryptocurrencies replace portfolios. FX volatility pricing persists even after controlling for established cryptocurrency return predictors (Lucey et al., 2022; Dobrynskaya, 2024; Lee and Wang, 2024) and volatility spillovers from stock and bond markets. These findings reaffirm the critical robust role of FX volatility in capturing cryptocurrencies' dynamic exposures to common sources of risk.

Building on the established pricing power of the FX volatility factor in cross-sectional settings, we extend our analysis to time-series contexts and document a significant negative spillover relationship from FX volatility to cryptocurrency returns. We test the profitability implications of this result using two time-series trading strategies that employ FX volatility as a cross-asset signal. Our first strategy follows the momentum framework, establishing long (short) positions in cryptocurrencies if their past weekly excess return is positive (negative) and that of the FX volatility is negative (positive). The second strategy employs moving-average (MA) technical trading signals, taking long (short) positions in a cryptocurrency when its short-term MA exceeds (falls below) its long-term MA. Both strategies corroborate our asset pricing results and document the economic success of our currency-based trading strategies, generating substantial statistical and economic profits across our cryptocurrency sample.

In a nutshell, FX volatility helps both price cryptocurrency returns and generate profits. Our time-series results show that FX volatility transmits systematic risk to crypto markets, while our cross-sectional results reveal that this monetary-competition risk is positively priced, so investors demand compensation for fiat-linked instability. Together, both results portray cryptocurrencies as imperfect monetary substitutes that inherit systemic fiat risks and are compensated for bearing them. We situate this new evidence within the theoretical framework of currency competition (Schilling and Uhlig, 2019; Benigno et al., 2022; Chiu and Koeppl, 2022), where cryptos and fiat currencies imperfectly substitute as stores of value, and fiat monetary uncertainty—proxied by FX volatility—emerges as a fundamental pricing factor.

From an academic perspective, we distinguish ourselves from the cryptocurrency literature by making the first attempt to explain the cross-section of cryptocurrency excess returns using cross-asset data from the FX market. Previous research focuses on explaining cross-sectional expected returns using cryptocurrency-specific factors, with an incomplete list of examples including market, size, and momentum (Liu et al., 2022), downside risk (Zhang et al., 2021; Dobrynskaya, 2024), realized volatility (Lee and Wang, 2024), liquidity (Borri and Shakhnov, 2022), uncertainty (Lucey et al., 2022), technical indicators (Fieberg et al., 2023), and network factors (Liu and Tsyvinski, 2021), among others. We acknowledge that there have been few attempts to relate cryptocurrencies to currencies within the framework of currency competition (Benigno et al., 2022), causality (Palazzi et al., 2021), and risk management (Ferroni, 2022), but none has done so in an asset pricing framework, to our knowledge.¹

From a practical perspective, our work offers practitioners key insights for cryptocurrency trading by demonstrating that incorporating historical FX volatility information improves cryptocurrency position management. We regard this latter evidence as being complementary to prior research on cross-asset trading strategies that combine signals from equity and bond markets (Pitkäjärvi et al., 2020), equity, bond, and FX markets (Phylaktis and Yamani, 2025; Goulding and Harvey, 2025), carry trade and commodities (Lu and Jacobsen, 2016), as well as crude oil volatility and stock markets (Fernandez-Perez et al., 2023).

The remainder of the paper proceeds as follows. Section 2 builds on the currency competition literature to present a stylized model linking FX volatility to cryptocurrencies. Section 3 documents the ability of FX volatility to price cryptocurrency returns, and Section 4 presents robustness tests. Section 5 documents the success of currency-based trading strategies in generating profits in the cryptocurrency market; Section 6 concludes.

¹Although several studies document the inter-relatedness of currencies and other asset classes such as stocks (Dobrynskaya, 2014), bonds (Ballensiefen and Ranaldo, 2023) as well as commodities and options (Kojen et al., 2018), the role of FX market information has surprisingly yet to be extended to cryptocurrencies.

2 Theoretical Framework

2.1 Motivation

The choice to use FX volatility risk to price cryptocurrency returns is inspired by an emerging literature that examines the connectedness between currencies and cryptocurrencies. From a theoretical viewpoint, the Crypto-Enforced Monetary Policy Synchronization (CEMPS) theory of [Benigno et al. \(2022\)](#) offers two channels for the currency-cryptocurrency relationship through the impact of currency competition between currencies and cryptocurrencies on interest rates and exchange rates (the first leg) as well as monetary policies (the second leg).

The first mechanism is driven by the classical literature on substitution theory and currency competition among fiat currencies ([Girton and Roper, 1981](#), [Kareken and Wallace, 1981](#), and [Matsuyama et al., 1993](#)) which has paved the way for a modern strand of literature that extends the analysis of currency competition to cryptocurrencies ([Schilling and Uhlig, 2019](#), [Fernández-Villaverde and Sanches, 2019](#), [Benigno et al., 2022](#), [Chiu and Koepl, 2022](#)). This line of research posits that private cryptocurrencies, unlinked from any sovereign entity, compete with local sovereign currencies, given that both currencies (physical and digital) are substitutes in providing liquidity services, thereby challenging national currency sovereignty.

[Benigno et al. \(2022\)](#), who examine currency substitution in a two-country model with complete markets featuring two national currencies and a global cryptocurrency that is used in both local markets, propose the CEMPS framework as a general stochastic structure through which cryptocurrencies affect interest rates and exchange rates. According to the CEPMS theory, the simultaneous circulation of global and local currencies requires the synchronization of monetary policies across countries, leading to equal national nominal interest rates.

To characterize the implications of the CEMPS for exchange rate dynamics, [Benigno et al. \(2022\)](#) argue that the two local currencies compete indirectly through the global currency, which enjoys worldwide acceptance and is used in both local markets, making the exchange rate between the national currencies a risk-adjusted martingale and inducing substantial exchange rate volatility. Along the same lines, [Schilling and Uhlig \(2019\)](#) analyze currency

competition between the US Dollar and the Bitcoin in an endowment economy, and show that Bitcoin, despite its price volatility, can be viewed as an international medium of exchange.

The second mechanism examines the implications of cryptocurrency competition for the international monetary system (Fernández-Villaverde and Sanches, 2019, Benigno, 2023, Benigno et al., 2022). It suggests that such competition constrains the central bank's ability to implement the monetary policy, including adjustments to benchmark interest rates and the money supply, thereby exerting significant effects on capital flows in cryptocurrency market.² Fernández-Villaverde and Sanches (2019), who build a model of currency competition among privately issued fiduciary currencies based on the framework of Lagos and Wright (2003), find that currency competition can lead to an inefficient social allocation of the money supply.

In this setup, cryptocurrencies limit central banks by constraining the effectiveness of their primary operational instrument – the interest rate (Benigno et al., 2022) and by restricting their ability to achieve core monetary policy objectives such as price stability and inflation control (Benigno, 2023). Under a currency substitution framework, central banks thus risk being trapped at the zero lower bound if they seek to lower the domestic interest rate in an attempt to preserve the role of the national currency as a medium of exchange by restricting the circulation of the global currency within the domestic economy.

2.2 Stylized Model

We capitalize on the currency competition theory (Von Hayek, 1976; Kareken and Wallace, 1981; Schilling and Uhlig, 2019) to present a model that links the stochastic discount factor (SDF) for cryptocurrency returns to FX volatility in an asset pricing framework. Consider a representative agent who allocates wealth W_t across two asset classes: (i) a fiat portfolio of sovereign currency bonds indexed by j , with gross nominal return $R_{j,t+1}$ and FX volatility exposure $\sigma_{FX,t}$; and (ii) a set of N cryptocurrencies indexed by i , serving as imperfect monetary substitutes for fiat money with returns $R_{i,t+1}$ and FX volatility exposure β_i^{FX} .

²For example, an increase in the U.S. interest rates may induce investors to reallocate capital from high-risk digital assets toward comparatively safer financial assets, such as U.S. government securities. Along the same lines, when the US dollar strengthens, cryptocurrency prices often fall, as the dollar's appreciation reduces the purchasing power of other currencies used to buy cryptocurrencies.

The agent maximizes lifetime utility from a composite consumption basket C_t denominated in U.S. dollars, following the standard [Cochrane \(2005\)](#) asset-pricing setting,

$$\max_{\{C_t, \omega_{i,t}\}} E_0 \left[\sum_{t=0}^{\infty} \beta^t \frac{C_t^{1-\gamma}}{1-\gamma} \right], \quad (1)$$

where $\beta \in (0, 1)$ represents the subjective discount factor and $\gamma > 1$ the coefficient of relative risk aversion. As in [Schilling and Uhlig \(2019\)](#), the agent faces a cash-in-advance (CIA) constraint that requires a fraction of consumption to be financed by monetary assets,

$$P_t C_t \leq M_t^{\text{fiat}} + \phi \sum_{i=1}^N q_{i,t} X_{i,t}, \quad (2)$$

where $q_{i,t}$ denotes the price of cryptocurrency i , $X_{i,t}$ its quantity held, and ϕ the substitutability parameter ([Kareken and Wallace, 1981](#)) governing crypto i 's substitution for fiat currency j .

The agent's first-order condition for any asset i yields the standard no-arbitrage restriction,

$$1 = E_t[M_{t+1} R_{i,t+1}], \quad (3)$$

where M_{t+1} denotes the SDF derived from the agent's first-order condition,

$$M_{t+1} = \beta \left(\frac{C_{t+1}}{C_t} \right)^{-\gamma} \cdot \Psi_{t+1}, \quad (4)$$

and Ψ_{t+1} represents the shadow price of the CIA constraint in Equation (2). By log-linearizing Equation (4) around the deterministic steady state ([Campbell and Shiller, 1988](#)) and substituting the consumption dynamics influenced by FX markets, we obtain

$$m_{t+1} \equiv \log M_{t+1} \approx -\gamma \Delta c_{t+1} - \theta \Delta \sigma_{FX,t+1} \quad (5)$$

where $\theta > 0$ denotes a reduced-form parameter increasing in ϕ , γ , and $\bar{\sigma}^2$.

Equation (5) implies that the SDF loads *negatively* on FX volatility ($-\theta \Delta \sigma_{FX,t+1}$), marking higher FX volatility as a bad state of the world ([Cochrane, 2005](#)) that raises M_{t+1} and lowers risky asset prices contemporaneously ([Menkhoff et al., 2012a](#)). From the standard

asset pricing relation (Cochrane, 2005), the expected excess return on cryptocurrency i is

$$E[r_i^e] = -\frac{\text{Cov}(m_{t+1}, r_{i,t+1}^e)}{E[M_{t+1}]} \quad (6)$$

Substituting the SDF (5) into Equation (6), we get

$$E[r_i^e] = \underbrace{\gamma \text{Cov}(\Delta c, r_i^e)}_{\text{consumption risk premium}} + \underbrace{\theta \text{Cov}(\Delta \sigma_{FX}, r_i^e)}_{\text{FX volatility risk premium}} \quad (7)$$

In currency competition equilibrium, the indifference condition (Schilling and Uhlig, 2019) requires cryptocurrency i 's expected return to equal fiat currency j 's return adjusted for the monetary risk premium, as follows

$$\underbrace{\frac{q_{i,t+1} - q_{i,t}}{q_{i,t}}}_{\text{crypto } i \text{ capital gain}} = \underbrace{r_{j,t}}_{\text{fiat } j \text{ return}} + \underbrace{\lambda \beta_i^{FX}}_{\text{monetary risk premium}} - \underbrace{\phi \mu_{i,t}}_{\text{monetary service yield}} \quad (8)$$

Equation (8) implies that crypto i must earn a higher expected capital gain than fiat j by the monetary risk premium $\lambda \beta_i^{FX}$, compensating for its monetary service yield $\phi \mu_{i,t}$. Within this setting, FX volatility thus acts as a wedge: higher $\sigma_{FX,t}$ tightens the equilibrium, depressing cryptocurrency i prices contemporaneously (time-series effect) while requiring a higher expected returns (cross-sectional effect). Since $\text{Cov}(\Delta \sigma_{FX}, r_i^e) = \beta_i^{FX} \cdot \text{Var}(\Delta \sigma_{FX}) < 0$ in Equation (7) (i.e., assets with positive β_i^{FX} co-move *negatively* with FX volatility, time-series effect), the cross-sectional pricing equation becomes

$$E[r_i^e] = \alpha + \underbrace{\theta \cdot \text{Var}(\Delta \sigma_{FX})}_{\lambda > 0} \cdot \beta_i^{FX} \quad (9)$$

Equation (9) constitutes the central result of the model: cryptocurrencies with higher FX volatility exposure β_i^{FX} earn higher average excess returns, implying that FX volatility carries a *positive* price of risk ($\lambda = \theta \cdot \text{Var}(\Delta \sigma_{FX}) > 0$) in the cryptocurrency excess return cross-section. This theoretical framework motivates our empirical analysis in the next section.

3 Does FX Volatility Price Cryptocurrency Risk?

3.1 Cryptocurrency Excess Returns

Our dataset comprises cryptocurrency data sourced from CoinCodex <https://coincodex.com>, alongside currency data obtained from DataStream. All variables are observed at a daily frequency but are aggregated to a weekly frequency. We consider “Monday” average-of-the-week values, given that there are 7 trading days per week for cryptocurrencies compared to 5 trading days per week for currencies. For notational clarity, cryptocurrencies are denoted by i , currencies by j , daily data by τ , and weekly data by t . Our sample period spans the weekly period from June 2, 2014 to October 28, 2024 – which corresponds to 544 weekly observations. The sample period begins at the earliest date when data are available for at least five cryptocurrencies in our sample, thereby enabling quintile-based portfolio sorting while accommodating variation in data availability across cryptocurrencies.

We analyze a sample comprising the 30 largest cryptocurrencies ranked by market capitalization. We apply a series of data filters to the initial sample, which starts with all cryptocurrencies having a market capitalization exceeding 1 billion dollars. We then filter out coins lacking complete data on price, volume, and market capitalization; those with fewer than 3 years of data (i.e., issued after October 2021); and Tether (USDT) due to its status as a stablecoin. Finally, we filter extreme outliers in cryptocurrency returns by winsorizing at the 0.5% and 99.5% percentiles, effectively removing the top and bottom 1% of observations.

Table 1 reports the mean and standard deviation of weekly cryptocurrency excess returns ($r_{i,t}$), dollar trading volume in billions ($V_{i,t}$), and market capitalization in trillions ($MV_{i,t}$) across the 30 cryptocurrencies over the full sample period. Excess returns are measured as the logarithmic changes in cryptocurrency prices minus the risk-free rate, proxied by the one-month Treasury bill rate, with Treasury bill data sourced from the Federal Reserve Economic Data (FRED) database. The results are also reported for a global cryptocurrency portfolio that assigns equal weights to all cryptocurrencies in our sample.

[Table 1 here.]

To account for systematic risk in the cryptocurrency market, we construct cryptocurrency market, size, and momentum risk factors following the three-factor model of [Liu et al. \(2022\)](#), our benchmark for cryptocurrency return cross-sections. The cryptocurrency market excess return, denoted as CMKT, is defined as the difference between the cryptocurrency market return—calculated as the equal-weighted return of all available constituent coins—and the risk-free rate. For the cryptocurrency size factor, each week we partition the 30 cryptocurrencies in our sample into five size groups based on market capitalization; the CSIZE factor is then measured as the return differential between the small- and large-cap portfolios. The cryptocurrency momentum factor is constructed analogously by sorting cryptocurrencies into five groups based on their lagged returns, with CMOM defined as the return difference between the high- and low-momentum portfolios.

3.2 FX Volatility Risk Factor

The FX volatility risk factor is constructed from information on WMR/Reuters spot exchange rates for the 10 major developed market currencies: Australian dollar (AUD), British pound (GBP), Canadian dollar (CAD), Danish krone (DKK), Euro (EUR), Japanese yen (JPY), New Zealand dollar (NZD), Norwegian krone (NOK), Swedish krona (SEK), and Swiss franc (SEK). All spot exchange rates are quoted from the point of view of a U.S. investor as the number of foreign currency units per one U.S. dollar.

We first construct a daily measure of global FX volatility, denoted as $fxvol_\tau$, by averaging the absolute daily log return for each currency i on day τ across all sample currencies available on that day. Our global FX volatility proxy in week t , denoted as $FXVOL_t$, is then given by average-of-the-week value of $fxvol_\tau$ across the 5 trading days per week, as follows

$$FXVOL_t = \frac{1}{T_t} \sum_{\tau \in T_t} fxvol_\tau = \frac{1}{T_t} \sum_{\tau \in T_t} \left[\sum_{j=1}^N \frac{|R_{j,\tau}|}{N_\tau} \right] \quad (10)$$

where $R_{j,\tau}$ denotes the log return of currency j on day τ , calculated as the logarithmic change in spot exchange rates, and N_τ represents the number of available currencies on day τ . This measure is similar in spirit to the FX volatility measure proposed by [Menkhoff et al. \(2012a\)](#).

To take a first look at the relation between FX volatility and cryptocurrency returns, the final two columns of Table 1 report the pairwise Pearson correlation coefficients (with associated p-values) between cryptocurrency excess returns and FX volatility, thereby assessing their average correlation over the full sample period. The results indicate that FX volatility exhibits a significantly negative correlation with cryptocurrency excess returns for 22 of the 30 sample coins, with correlation coefficients ranging from -0.088 (for Cardano) to -0.167 (for Algorand). The last three rows of Table 1 report pairwise Pearson correlation coefficients between cryptocurrency risk factors (CMKT, CSIZE, and CMOM) and FXVOL factor.

3.3 FX Volatility and the Cross-Section of Cryptocurrency Returns

Our goal is to examine whether FX volatility, as proxied in Equation 10, is priced in cryptocurrency excess returns for five single-sorted portfolios formed by sorting cryptocurrencies on their lagged returns, which serve as our test assets. We test the cross-sectional implications of this hypothesis by estimating a linear four-factor model comprising FXVOL, CMKT, CSIZE, and CMOM, utilizing the standard two-pass regression methodology of Fama and MacBeth (1973) (FMB). In the first pass, we estimate time-series regressions of cryptocurrency portfolio excess returns $r_{p,t}$ for each quintile portfolio p in week t on the risk factors,

$$r_{p,t+1} = \alpha_i + \beta_{p,1}FXVOL_{t+1} + \beta_{p,2}CMKT_{t+1} + \beta_{p,3}CSIZE_{t+1} + \beta_{p,4}CMOM_{t+1} + \xi_{p,t+1}. \quad (11)$$

In the second pass, we run a cross sectional regression of average excess returns on the estimated betas from the first stage to estimate the prices of risk,

$$\bar{r}_{p,t} = \hat{\beta}_{1p}\lambda_{FXVOL} + \hat{\beta}_{2p}\lambda_{CMKT} + \hat{\beta}_{3p}\lambda_{CSIZE} + \hat{\beta}_{4p}\lambda_{CMOM} + \xi_t. \quad (12)$$

Subsequently, we refer to this four-factor model (as defined in Equations 11 and 12) as our baseline model that relies on the notion that cryptocurrency excess returns are exposed to systematic risk in the FX market (proxied by the FXVOL factor that extracts the global component in the FX market) that is additional to the information in systematic risk in the cryptocurrency market (proxied by CMKT, CSIZE, and CMOM factors in Liu et al., 2022).

Panel A of Table 2 presents the cross-sectional asset pricing results from the second pass of the FMB procedure (i.e., the estimated cross-sectional lambdas λ s in Equation 12, along with their associated Newey and West (1986) standard errors), where the test assets consist of average cryptocurrency excess returns on five single-sorted return portfolios formed by sorting cryptocurrencies on their lagged returns. We also report the cross-sectional model fit statistics – the Hansen and Jagannathan (1997) (HJ) for the null hypothesis that pricing errors are zero, the root mean squared pricing errors (RMSE), and the R-squared statistics.

We are primarily interested in the estimated price of risk associated with the FXVOL factor, which is shown to be positive, equal to 4.4% per week, and statistically significant at the 1% level, even after controlling for the benchmark cryptocurrency return determinants (CMKT, CSIZE, and CMOM) in Liu et al. (2022). This empirical evidence validates our theoretical prediction in Equation (9), which implies that FX volatility commands a positive price of risk in the cross-section of cryptocurrency returns. Our baseline model yields a high R-squared of about 96.4% and we cannot reject the null that the pricing errors are zero.

For completeness, Table 2 reports results for four additional quintile portfolios: (1) volume portfolios sorted on lagged trading volume (Panel B), (2) size portfolios sorted on lagged market value (Panel C), (3) volatility portfolios sorted on lagged realized volatility (Panel D), and (4) liquidity portfolios sorted on lagged liquidity. We construct a daily realized volatility measure, defined as $cvol_{i,\tau} = |R_{i,\tau}|$, where $R_{i,\tau}$ denotes the log return of cryptocurrency i on day τ . We construct an analogous daily liquidity measure for each cryptocurrency, defined as $cliq_{i,\tau} = \frac{|R_{i,\tau}|}{V_{i,\tau}}$, where $V_{i,\tau}$ denotes the trading volume of cryptocurrency i on day τ . The weekly measures are then given by average-of-the-week values of $cvol_{i,\tau}$ and $cliq_{i,\tau}$.

Looking at the overall results, it can be seen that the findings are robust to the choice of portfolio construction sorting variable. We find that the pricing power of the FXVOL factor remains significant at the 1% level across the different variants of cryptocurrency portfolios, ranging from 0.7% for the size portfolios to 5.6% for the volume and volatility portfolios, suggesting that FX volatility is successful in pricing cryptocurrency excess returns.

[Table 2 here.]

4 Robustness Tests

In this section, we explore several robustness checks on our evidence regarding the power of FX volatility risk in pricing cryptocurrency returns, as shown in Table 2. To scrutinize the robustness of this pricing power, we assess various specifications of our baseline asset pricing model using alternative measures for cryptocurrency returns, FXVOL factor, sample period, and test assets. We also examine whether the pricing power of the FXVOL factor is driven by systematic risk in the cryptocurrency market or other financial markets.

4.1 Value-Weighted Returns

To assess the robustness of cryptocurrency return measurement, Table 3 repeats the same portfolio-level asset pricing analysis in Table 2 with the only exception that our test assets now comprise the value-weighted (in lieu of equally weighted) excess returns on five cryptocurrency portfolios constructed by sorting currencies based on their lagged return, volume, size, volatility, and liquidity. We first calculate weights for each cryptocurrency i on each day τ in our sample by dividing each crypto's market cap by the total market cap. We then construct a daily measure for value-weighted returns by multiplying each crypto's return by its weight. The weekly value-weighted returns are measured as the average of the daily returns for the seven days. Examining the overall value-weighted results in Table 3, all five portfolios show consistent results. The risk price estimates for the FXVOL factor consistently exhibit statistical significance and positive values across all portfolio variants, indicating that our findings remain robust across alternative cryptocurrency return measurement methods.

[Table 3 here.]

4.2 Emerging Market Currencies

We next examine whether our baseline results in Table 2 are driven by our sample choice of the currencies used in constructing the FXVOL factor. For robustness, we extend our original sample of 10 developed market currencies to encompass 12 additional emerging

market currencies: Chilean Peso (CLP), Czech Koruna (CZE), Hungarian Forint (HUN), Indian Rupee (INDI), Indonesian Rupiah (INDO), Malaysian Ringgit (MAL), Mexican Peso (MEX), Peruvian Sol (PEN), South African Rand (SAF), Singapore Dollar (SIN), South Korean Won (KRW), and Thai Baht (THB). Table 4 presents the results for this analysis.

The setup in Table 4 is the same as in Table 2 except that we now use developed and emerging market currencies to construct the FXVOL factor. Compared to Table 2, we observe that all results remain virtually unchanged in Table 4. The lambda estimates for the FXVOL factor continue to be positive and statistically significant across the different variants of cryptocurrency portfolios. Therefore, our findings remain robust whether we consider only developed countries or the combined sample of developed and emerging countries.

[Table 4 here.]

4.3 Subsample Analysis

We further investigate whether time-varying risk and return dynamics, driven by evolving market conditions, affect the pricing power of the FXVOL factor in cryptocurrency returns. To this end, Table 5 presents the cross-sectional asset pricing results for our baseline model, estimated over four subsample periods centered around the COVID-19 period: (1) Pre-COVID period: June 2, 2014, to January 31, 2020 (Panel A); (2) Post-COVID period: February 1, 2020, to October 31, 2024 (Panel B); (3) Bull market period: January 1, 2020, to November 14, 2021 (Panel C); and (4) Bear market period: November 15, 2021, to October 31, 2024 (Panel D). The test assets are the average equally weighted excess returns of quintile cryptocurrency portfolios sorted on the prior week cryptocurrency excess return. Our subsample analysis in Table 5 corroborates the full-sample findings, indicating that the FXVOL factor exerts a significant and robust effect on cryptocurrency returns across tranquil and turmoil periods.

[Table 5 here.]

4.4 Crypto-level Asset Pricing

As an alternative approach to assessing the importance of FX volatility risk in cryptocurrency returns, we conduct a crypto-level analysis to mitigate concerns regarding data-snooping biases or the use of too few portfolios as test assets (Lo and MacKinlay, 1990 and Lewellen et al., 2010). In Table 6, we extend our asset pricing tests to individual cryptocurrencies, comparing the results with those from cryptocurrency portfolios (Table 2) as two prominent performance criteria in the asset pricing literature. For robustness, Table 6 reports the crypto-level asset pricing results using cryptocurrency samples of varying sizes: specifically, the 30 (Panel A), 20 (Panel B), 10 (Panel C), and 5 (Panel D) largest cryptocurrencies ranked by market capitalization. Overall, the pricing power of the FXVOL factor is maintained when individual cryptocurrencies are used as test assets, given that the FXVOL factor continues to be positive and statistically significant at 1% level across the four variants of samples. This evidence underscores the pivotal role of the FXVOL factor in capturing cryptocurrencies' exposures to common sources of risk at both the individual-crypto and portfolio levels.

[Table 6 here.]

4.5 Global Risk Factors from the Cryptocurrency Market

Our baseline model in Equation 12 uses the market, size, and momentum factors in Liu et al. (2022) to account for systematic risk in the cryptocurrency market. Nevertheless, the growing cryptocurrency literature suggests other cryptocurrency-specific risk factors, such as (1) the downside market risk factor (CMKTDR) of Zhang et al. (2021) and Dobrynskaya (2024); (2) the volatility factor (CVOL) of Lee and Wang (2024); (3) the liquidity factor (CLIQ) of Brauneis et al. (2021); (4) the uncertainty index (UCRY) of Lucey et al. (2022), that captures the uncertainty risk in the cryptocurrency market; and, (5) the geopolitical risk index (GPR) of Caldara and Iacoviello (2022), that quantifies risks associated with major adverse geopolitical events. Therefore, our evidence on the FXVOL factor's success in pricing cryptocurrency returns may be driven by omitted factors related to unexpected changes in cryptocurrency market conditions, such as price volatility, liquidity, and geopolitical risks.

To address this concern, we subject our analysis to a more stringent test by augmenting our baseline model in Equation 12 to separately control for CMKTDR, CVOL, CLIQ, UCRY, and GPR. Following Dobrynskaya (2024), CMKTDR is defined as an interaction dummy variable ($\text{CMKTDR} = \text{CMKT} \times \text{Dummy}$) that takes the value of zero for negative CMKT values and one otherwise in order to control for adverse financial market conditions.

Both CVOL and CLIQ measures are constructed in a similar spirit to the FXVOL measure. We first calculate a daily measure for global cryptocurrency volatility, $\text{cvol}_\tau = \sum_{i=1}^N (\frac{|R_{i,\tau}|}{N_\tau})$, where $R_{i,\tau}$ denotes the log return of cryptocurrency i on day τ , and N_τ represents the number of available cryptocurrencies on day τ . Analogously, we follow Brauneis et al. (2021) and construct a daily measure for the Amihud (2002) liquidity factor, $\text{cliq}_\tau = \sum_{i=1}^N (\frac{|R_{i,\tau}|}{V_{i,\tau}})$, where $V_{i,\tau}$ denotes the trading volume of cryptocurrency i on day τ . The weekly measures of volatility (CVOL_t) and liquidity (CLIQ_t) are then given by the averages of the daily values of cvol_τ and cliq_τ , respectively, to capture volatility and liquidity risks in the crypto market.

To assess the relative pricing power of the aforementioned five risk factors (CMKTDR, CVOL, CLIQ, UCRY, and GPR)³ relative to the FXVOL factor, we conduct a horse-race test between these five control factors and the FXVOL factor using two alternative model specifications. The first specification excludes FXVOL, whereas the second includes it, while controlling for CMKT, CSIZE, and CMOM in both model specifications. Therefore, this analysis includes 10 regressions. Table 7 reports the asset pricing results for this analysis.

Finding that the inclusion of any of these control factors eliminates the statistical significance of the FXVOL factor would imply that its pricing power is driven by these factors. Examining the results in Table 7, we observe that the opposite is the case: the FXVOL factor eliminates the statistical significance of these cryptocurrency risk factors. In the first specification, which excludes FXVOL, the lambda estimates for all five factors are statistically significant—a finding consistent with prior research (Lee and Wang, 2024; Brauneis et al., 2021; Lucey et al., 2022; Caldara and Iacoviello, 2022).

³We obtain the cryptocurrency uncertainty index (UCRY) data from the digital asset indices website: <https://sites.google.com/view/cryptocurrency-indices/the-indices/crypto-uncertainty?authuser=0>. GPR data are obtained from the website of Matteo Iacoviello: <https://www.matteoiacoviello.com/gpr.htm>.

Once the FXVOL factor is accounted for in the second specification, the conclusions regarding the significance of the pricing role of the five control factors in cryptocurrency returns are reversed. Specifically, the FXVOL factor remains significantly positive across all the different models, with no additional explanatory power from the control factors—with the sole exception of the CMKTDR factor. This latter finding lends credence to the pricing of downside risk in cryptocurrency returns (Zhang et al., 2021; Dobrynskaya, 2024).

[Table 7 here.]

4.6 Global Risk Factors from Other Financial Markets

Our overall results show that the FX volatility risk is priced in the cross section of cryptocurrency excess returns. This raises the question of whether other risk factors—beyond FXVOL and constructed from volatility information in alternative financial markets—can also help in pricing cryptocurrency returns. To address this question, we construct equity market volatility (EQVOL) and sovereign bond market volatility (BOVOL) risk factors. We first construct a daily measure of global equity volatility, denoted $eqvol_\tau$, by averaging the absolute daily log equity returns for each country j on each day τ across all countries on any given day. The weekly global equity market volatility proxy, $EQVOL_t$, is then given by average-of-the week value of $eqvol_\tau$ across the 5 trading days per week. The $BOVOL_t$ factor is constructed in a similar fashion, using quotes for sovereign bond indices.

Table 8 reports the results for six different specifications of a five-factor model based on Equation 12 augmented with one of the following variables: (1) EQVOL, (2) BOVOL, (3) the MSCI world stock market index return (WOEQ), (4) the Citibank world government bond index return (WOBO), (5) the US Datastream equity index return (USQEQ), (6) the US 5-year Datastream government bond index return (USBO), and (7) the change in the VIX index.⁴ Our goal is to assess whether the pricing power of the FXVOL factor can be explained by relying on its covariance with variables from equity and bond markets.

⁴We construct EQVOL and BOVOL factors using data for Datastream stock market indices and 5-year Datastream government bond indices for our sample of 10 developed countries (Australia, Canada, Germany, Japan, New Zealand, Norway, Switzerland, Sweden, Denmark and the UK). Data for stock and bond returns are obtained from Datastream, while data for the VIX index are obtained from the FRED database website.

Overall, the evidence in Table 8 continues to underscore the important role of FX volatility in capturing cryptocurrencies' exposures to common sources of risk. Table 8 shows that FXVOL estimates remain consistently positive and statistically significant in the presence of equity and bond variables. Interestingly, even after controlling for the CMKT, CSIZE, and CMOM factors, both the EQVOL and BOVOL factors bear significantly positive risk prices of 5.5% and 1.2%, respectively, both significant at the 1% level. These findings suggest that volatility information from FX, equity, and bond markets helps in pricing cryptocurrency returns, indicating that cryptocurrencies are not fully insulated from traditional financial system shocks, with their risk premiums influenced by global foreign exchange stress.

[Table 8 here.]

4.7 Systematic Risk Spillovers

To better elucidate the connection between cryptocurrencies and fiat currencies, we investigate the extent to which established return predictors in FX markets are reflected in cryptocurrency returns, and conversely, whether predictors from cryptocurrency markets are reflected in currency returns. To this end, Table 9 presents a comparative asset-pricing analysis of cryptocurrency returns (Panel A) and currency returns (Panel B) with respect to prominent determinants of returns in the FX and cryptocurrency markets, respectively.

Panel A of Table 9 reports asset pricing results for an alternative specification of our baseline model, replacing FXVOL with three prominent currency-based risk factors: the dollar (DOL) factor, the carry (CAR) factor, and the currency momentum (MOM) factor, all of which have been shown to price currency excess returns in the FX market literature (Lustig et al., 2011 and Menkhoff et al., 2012b).⁵ Panel B of Table 9 repeats the same analysis as in Panel A except that now we use quintile currency portfolios (in lieu of cryptocurrency ones) sorted based on the prior week currency excess return as test assets and the DOL, CAR, and

⁵The dollar factor is defined as the average currency excess returns of five single-sorted forward discount portfolios. The carry factor is measured as the currency return differential between the portfolios with the highest and lowest forward discount. The currency momentum factor is calculated as the currency return differential between the portfolios with the highest and lowest lagged currency return.

MOM factors (in lieu of the CMKT, CSIZE, and CMOM factors). For comparison purposes, we also report the response of both test assets to FXVOL, CVOL, and CLIQ factors.

Two interesting findings emerge from the comparative analysis reported in Table 9, relating to both the statistical significance and the signs of the estimated risk prices. First, cryptocurrency returns are exposed to systematic risk originating in the FX market that is distinct from and additional to the information conveyed by systematic risk in the cryptocurrency market, and vice versa. Specifically, Panel A reveals that not only the FXVOL factor but also both the CAR and MOM factors are priced in the cross-section of cryptocurrency returns, even in the presence of popular determinants of cryptocurrency returns. In comparison, Panel B shows that the estimated risk prices of CMKT, CSIZE, and CMOM are statistically significant at the 1% level when evaluated with respect to currency portfolios, even after controlling for systematic risk in the FX market (proxied by DOL, CAR, and MOM factors). Analogously, the volatility risks embedded in both the cryptocurrency and foreign exchange markets (CVOL and FXVOL) are priced across both markets.

Second, we capture asymmetric responses to currency-based factors across cryptocurrency and currency portfolios in several instances. Put differently, many of our estimated risk prices flip sign when switching from cryptocurrency portfolios to currency portfolios as test assets. For example, we find that cryptocurrency excess returns display a significantly negative sensitivity to both CAR and MOM factors in Panel A of Table 9. In contrast, currency excess returns display a consistently significant and positive pricing relation with both variables across all model specifications in Panel B of Table 9, a finding which is consistent with [Lustig et al. \(2011\)](#) and [Menkhoff et al. \(2012b\)](#). These sign flips are also observed for the FXVOL factor, which is our main variable of interest. Our results indicate that FXVOL commands a significantly positive cross-sectional price of risk with respect to cryptocurrency portfolios, whereas its estimated risk price becomes significantly negative when currency portfolios are considered. This latter finding is in line with [Menkhoff et al. \(2012a\)](#), implying common systematic factors underpin cross-sectional pricing across crypto and fiat currency markets.

[Table 9 here.]

In sum, our empirical findings point to the novel conclusion that FX volatility matters for understanding the cross-section of cryptocurrency returns (Table 2). This pricing power remains robust even when (1) value-weighted returns on cryptocurrencies (in place of equally-weighted returns) are used as test assets in Table 3; (2) both emerging- and developed-market currencies (rather than developed-market currencies only) are employed to construct the FXVOL factor in Table 4; (3) subsample periods (instead of the full sample period) are considered in Table 5; and (4) individual cryptocurrencies (rather than cross-sectional portfolios) are used as test assets in Table 6. The pricing power of FX volatility is also not driven by standard determinants of cryptocurrency returns (Table 7) and is distinct from equity and bond volatilities (Table 8). Furthermore, we find that systematic risk factors in the cryptocurrency market affect currencies and vice versa, highlighting a two-way dependence in their cross-sectional pricing and shared risk dynamics (Table 9).

This established cross-market nexus, captured in Sections 3 and 4, operationalizes the stylized model outlined in Section 2 and can be understood within the theoretical framework of currency-competition between traditional fiat currencies and cryptocurrencies (Schilling and Uhlig, 2019, Benigno et al., 2022, Chiu and Koepl, 2022), particularly with respect to interest rates and exchange rates. Within this framework, cryptocurrencies, when used alongside national currencies, act as a disciplinary force on monetary policy, inducing pressure toward interest-rate synchronization and, in some cases, amplifying exchange-rate volatility and financial-stability risks (Kroszner 2007). The shared risk factors arising from currency-substitution pressures help explain the joint cross-sectional pricing dynamics between cryptocurrencies and sovereign currencies amid intensifying policy-convergence pressures.

In such a setting, the significant premium on FX volatility, viewed as a proxy for global macroeconomic uncertainty, implies that crypto investors are compensated for bearing exposure to this global FX-related risk. Within a standard risk–return framework, the positive price of FX volatility implies that higher FX-volatility exposure is associated with significantly higher expected cryptocurrency returns, consistent with the presence of a significant risk premium in the cross-section of cryptocurrency returns.

5 Cross-Asset Trading Strategies: Time-Series View

In this section, we move beyond cross-sectional asset-pricing analysis and adopt a time-series, practical perspective to study whether FX volatility can be used as a trading signal to construct profitable cryptocurrency strategies. Despite its importance, the cryptocurrency literature offers no evidence on the economic value of currency-based signals. We take a step in that direction by developing cross-asset trading strategies to study how much crypto investors can gain from trading conditional on FX-market cross-asset signals. Our approach complements prior work on cross-asset trading strategies that combine information from stock and bond markets (Pitkäjärvi et al., 2020), carry trade and commodity markets (Lu and Jacobsen, 2016), and crude oil volatility and equity markets (Fernandez-Perez et al., 2023).

5.1 The Cross-Asset Spillover Effects of FX Volatility

To set the stage for developing cross-asset trading strategies, we first assess the significance and direction of time-series linkages in FXVOL spillovers between the cryptocurrency and FX markets. To this end, we use a vector autoregressive model with error terms following a GARCH-in-mean process, a model known as the VAR-GARCH-M (Engle and Kroner, 1995),

$$\begin{cases} Y_t = M \cdot FXVOL_t + A \cdot Factors_t + R \cdot Y_{t-1} + \varepsilon_t \\ H_t = Z' r_{t-1} r'_{t-1} Z + G' H_{t-1} H'_{t-1} G \end{cases} \quad (13)$$

where

$$Y_t = \begin{bmatrix} r_{p,t}^{Crypto} \\ r_{p,t}^{FX} \end{bmatrix}, \quad M = \begin{bmatrix} m_{11} \\ m_{21} \end{bmatrix}, \quad A = \begin{bmatrix} a_{11} \\ a_{21} \end{bmatrix}, \quad R = \begin{bmatrix} r_{11} & r_{12} \\ r_{21} & r_{22} \end{bmatrix}$$

Equation 13 allows us to analyze the simultaneous response of cryptocurrency returns ($r_{p,t}^{Crypto}$) and currency returns ($r_{p,t}^{FX}$)⁶ to FXVOL spillovers through the M -matrix elements m_{11} and m_{21} , which measure the spillover to cryptocurrency and currency returns, respectively.

⁶The equally weighted cryptocurrency portfolio combines individual cryptocurrencies to yield a return calculated simply as average returns on the individual cryptocurrencies: $r_{p,t}^{Crypto} = \frac{1}{N} \sum_{i=1}^N (R_{i,t})$, where N is the set of available cryptocurrencies at week t . Analogously, the equally weighted currency portfolio return is defined as average returns on the individual 10 developed market currencies: $r_{p,t}^{FX} = \frac{1}{10} \sum_{j=1}^{10} (R_{j,t})$.

To control for potential cross-asset return linkages between the two markets, we include two additional interaction channels. The A -matrix captures the joint impact of cryptocurrency risk factors (CMKT, CSIZE, CMOM) on both portfolio returns, while the R -matrix captures cross-market return spillovers and autoregressive dynamics. The off-diagonal elements r_{12} and r_{21} of the R -matrix capture how cryptocurrency and currency returns respond to each other's lagged returns, while the diagonal elements r_{11} and r_{22} capture their responses to their own lags. The Z and G -matrices encode the GARCH parameters.

Table 10 reports the VAR-GARCH-M estimates for the elements of the M , A , and R matrices, using $r_{p,t}^{Crypto}$ (Panel A) and $r_{p,t}^{FX}$ (Panel B) as test assets. To assess the importance of cross-market effects in our analysis, we estimate three different specifications of Equation 13 that vary in the set of control variables included with the FXVOL factor. The first specification includes only the cryptocurrency risk factors (i.e., the elements of the A -matrix), the second includes the lagged crypto and currency returns (i.e., the elements of the R -matrix), and the third includes both. Consequently, this analysis consists of six regressions.

Our primary interest is to examine the significance and direction of FX volatility spillovers on cryptocurrency returns. In Panel A of Table 10, we find a significant negative impact of FX volatility on cryptocurrency returns across all model specifications, lending credence to our theoretical framework in Equation 8 which implies that a rise in FX volatility depresses cryptocurrency prices contemporaneously. This evidence is further supported by the pairwise Pearson correlations in Table 1, which show a negative correlation of -0.11 between the cryptocurrency portfolio return and FX volatility, significant at the 5% level.

A contrasting pattern emerges when we examine the response of currency returns to FX volatility in Panel B of Table 10. Interestingly, the relationship between the two variables turns out to be insignificant once cross-market effects are taken into account. The cross-market return spillover results suggest that there is a unidirectional positive return spillover from cryptocurrency markets to currency markets (given by significant estimates of r_{21}), suggesting that currency returns react positively to cryptocurrency returns in the previous period.

[Table 10 here.]

5.2 Cross-Asset Time-Series Momentum

Motivated by our evidence of a significant time-series relationship between cryptocurrency returns and FXVOL in Table 10, we design a cross-asset time-series trading strategy within the momentum framework. The single-asset time-series momentum has been widely used in the literature on stocks, bonds, currencies (Moskowitz et al., 2012), commodities (Georgopoulou and Wang, 2017), and futures (Kim et al., 2016), where investors use a single-asset predictor to go long (short) winner (loser) assets, betting on return continuation. We extend earlier research on single-asset cryptocurrency momentum (Liu et al., 2022; Fieberg et al., 2023) and design a cross-asset momentum trading strategy using past cryptocurrency returns (same-asset signal) and FX volatility (cross-asset signal) as joint predictors.

We start with the construction of weekly trading signals, denoted as $Z_{i,t}^{TSMOM}$, for every single cryptocurrency i as follows. We assume that traders take a long position in cryptocurrency i (i.e., $Z_{i,t}^{TSMOM} = +1$) if the past weekly cryptocurrency excess return is positive and that of the FX volatility is negative. Conversely, they will take a short position in cryptocurrency i (i.e., $Z_{i,t}^{TSMOM} = -1$) if the past weekly cryptocurrency excess return is negative and that of the FX volatility is positive. Otherwise, traders take no position. Given such a setup, the excess return on our cross-asset time series momentum strategy, denoted as $r_{i,t}^{TSMOM}$, equals the actual crypto return multiplied by the trading signal, as follows

$$r_{i,t}^{TSMOM} = \begin{cases} +1 \times r_{i,t}, & \text{if } r_{i,t-w} > 0 \text{ and } FXVOL_{t-w} < 0. \\ -1 \times r_{i,t}, & \text{if } r_{i,t-w} < 0 \text{ and } FXVOL_{t-w} > 0. \\ 0, & \text{otherwise.} \end{cases} \quad (14)$$

Table 11 reports the economic evaluation results for this trading strategy, using three weekly performance measures – mean, volatility, and skewness. Panel A of Table 11 reports these statistics for the five largest individual cryptocurrencies by market capitalization – Bitcoin, Ethereum, XRP, Binance, and Solana. For robustness, we investigate the profitability of four momentum indicators corresponding to lag lengths of $w = 10, 20, 50$ and 100 which we refer to as MOM(10), MOM(20), MOM(50) and MOM(100). Panel B of Table 11 reports

results for an equally weighted portfolio that combines individual cryptocurrencies to yield a return calculated simply as average returns on individual cryptocurrencies: $TSMOM_t = \frac{1}{N} \sum_{i=1}^N (TSMOM_{i,t})$, where N is the set of available cryptocurrencies at week t .

The overall results show that our cross-asset momentum strategy performs well. Namely, trading based on the sign of the MOM(10) signal in Bitcoin generates positive and statistically significant returns of 7.01% (compared to 6.85% for Ethereum and 6.89% for XRP). This evidence is also supported for the cryptocurrency cross-sectional portfolio, which yields a high and statistically significant return of 6.96%. Similar results are observed for both medium- and long-term momentum strategies using lags of 20, 50 and 100 weeks.

[Table 11 here.]

5.3 Cross-Asset Time-Series Moving Average

For robustness, we design an alternative time-series trading strategy similar to the one in the previous section, except that the cross-asset trading signal is now based on an MA technical trading rule. We first compute a weekly moving average of the FXVOL factor, $MA_t^{FXVOL} = \frac{1}{j} \sum_{w=0}^{j-1} FXVOL_{t-w}$ for $j = s, l$, where s (l) is the length of the short (long) MA ($s < l$). The MA-based long-short signals are defined for each cryptocurrency i as

$$Z_{i,t}^{TSMA} = \begin{cases} 1, & \text{if } MA_{s,t}^{FXVOL} \geq MA_{l,t}^{FXVOL}, \\ -1, & \text{if } MA_{s,t}^{FXVOL} < MA_{l,t}^{FXVOL}. \end{cases} \quad (15)$$

where we take a long position in a given cryptocurrency in a given week if the short moving average $MA_{s,t}$ is greater than or equal the long moving average $MA_{l,t}$. Otherwise, we take a short position. The excess returns, $r_{i,t}^{TSMA}$, for cryptocurrency i in week t are obtained as

$$r_{i,t}^{TSMA} = \begin{cases} +1 \times r_{i,t}, & \text{if } r_{i,t-w} > 0 \text{ and } Z_{i,t}^{TSMA} = 1. \\ -1 \times r_{i,t}, & \text{if } r_{i,t-w} < 0 \text{ and } Z_{i,t}^{TSMA} = -1. \\ 0, & \text{otherwise.} \end{cases} \quad (16)$$

Table 12 presents the profitability evaluation results for this MA-based trading strategy.

To examine whether any predictive effect disappears after some delay, we employ four weekly MA rules: (10,50), (10,100), (20,50), and (20,100), all of which provide consistent results, suggesting that our MA-based strategy yields economically meaningful profits. Panel A shows that the long–short MA payoffs for all the top five cryptocurrencies are significantly positive across the four MA trading rules. Similar results are observed for the cross-sectional portfolio in Panel B, where MA(10,50) and MA(10,100) yield significant positive weekly returns of 4.68% and 4.96%, respectively (compared to 4.74% and 4.86% for the 20-week MA).

[Table 12 here.]

6 Conclusion

In this study, we document that FX volatility is an empirically powerful variable in explaining cryptocurrency returns from both asset-pricing and trading-strategy perspectives. From an asset-pricing standpoint, we show that FX volatility is positively priced in the cross-section of cryptocurrency returns, suggesting that cryptocurrencies with greater exposure to FX volatility earn higher average returns, as investors demand a premium for holding assets whose payoffs co-move with FX volatility under a standard risk-adjustment framework. This pricing power is robust to various model specifications and empirical approaches.

From a trading-strategy perspective, we find that cryptocurrency excess returns fall on average when FX volatility rises contemporaneously in a time-series context. To highlight the practical implications of this result, we construct hypothetical long–short trading strategies in which investors take a long (short) position in cryptocurrencies when past cryptocurrency excess returns are positive (negative) and past FX volatility is negative (positive). The results document the economic success of these cross-asset trading strategies in generating both high statistical significance and economically positive returns for our sample of cryptocurrencies.

Overall, we establish FX volatility as a bona fide priced risk factor in cryptocurrency markets: it depresses returns in the time-series, yet the cross-section rewards higher exposure with greater average returns, implying that the negative contemporaneous reaction when

FX volatility rises reflects a systematic risk that justifies a positive risk premium—the cross-section compensates investors for the shock, while the time-series records it.

From a theoretical viewpoint, this cross-market nexus between traditional currencies and cryptocurrencies is framed well within the currency competition literature ([Schilling and Uhlig, 2019](#); [Benigno et al., 2022](#); [Chiu and Koepl, 2022](#)), which posits both as substitutes for liquidity services that foster competition undermining national monetary control. This framework forces monetary policy convergence but generates FX volatility and financial instability risks, suggesting shared risk factors shape cross-sectional pricing of cryptocurrencies and currencies, highlighting their mutual sensitivity and interconnected risk dynamics. Our evidence—of a positive price of risk coupled with a negative contemporaneous impact—can be reconciled within several channels, including the intertemporal hedging channel ([Merton, 1973](#)), the Campbell–Shiller discount-rate decomposition channel ([Campbell and Shiller, 1988](#)), and the flight-to-safety or liquidity channel ([Anadu et al., 2025](#)).

This study should inspire further research in the cryptocurrency asset-pricing literature and in real-world cryptocurrency trading, as our findings indicate that FX volatility plays an important role in pricing cryptocurrency returns and in creating profitable trading opportunities. From an academic standpoint, our work should motivate future research that deepens the understanding of the drivers of the FX volatility factor and its relationship with cryptocurrency risk premia. From a practical perspective, our findings should encourage cryptocurrency traders to refine their position management by incorporating the informational content of historical FX volatility into their cryptocurrency return forecasts.

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Table 1: Descriptive Statistics

Cryptocurrency variable	Crypto symbol	$R_{i,t}$		$V_{i,t}$		$MV_{i,t}$		$FXVOL_t$	
		Mean	SD	Mean	SD	Mean	SD	correl.	Pvalue
Bitcoin	BTC	-7.014	1.372	2215.425	306.668	2537.140	194.001	-0.089	0.038
Ethereum	ETH	-6.909	2.052	2164.535	311.054	2423.119	237.476	-0.072	0.115
Ripple	XRP	-7.028	2.262	1902.048	366.956	2244.214	210.306	-0.096	0.026
Binance	BNB	-7.023	1.875	2040.230	168.054	2332.289	174.736	-0.107	0.049
Solana	SOL	-6.758	2.464	2090.127	192.689	2284.006	221.576	-0.092	0.164
Tron	TRX	-7.058	2.083	2011.322	145.357	2192.321	97.148	-0.110	0.034
Dogecoin	DOGE	-6.995	2.383	1730.505	392.116	2020.789	270.273	-0.088	0.041
Cardano	ADA	-7.071	2.143	1970.212	159.143	2269.695	129.796	-0.088	0.092
Bitcoin Cash	BCH	-7.119	2.437	2044.669	95.115	2251.521	67.088	-0.130	0.012
Monero	XMR	-7.028	2.230	1683.746	322.393	2003.881	253.220	-0.076	0.077
Chain Link	LINK	-6.984	2.424	1912.343	246.531	2133.549	167.356	-0.136	0.009
UNUS SED LEO	LEO	-7.055	0.864	1464.112	114.765	2167.048	61.606	-0.037	0.537
Stellar	XLM	-7.070	2.298	1656.810	385.599	2022.473	255.939	-0.060	0.166
Zcash	ZEC	-7.211	2.355	1844.499	150.692	2009.973	107.557	-0.126	0.010
Avalanche	AVAX	-7.012	2.670	2000.050	111.420	2239.902	126.226	-0.115	0.092
Litecoin	LTC	-7.089	1.874	1917.019	290.392	2137.434	167.540	-0.090	0.037
Hedera	HBAR	-7.134	2.007	1769.736	129.098	2046.516	147.581	-0.141	0.022
Cronos	CRO	-7.135	2.059	1670.670	142.105	2122.925	120.900	-0.113	0.049
Polkadot	DOT	-7.150	1.933	2032.333	81.526	2292.043	69.270	-0.092	0.177
Uni Swap	UNI	-7.128	2.113	1979.891	65.733	2223.720	78.885	-0.123	0.071
AAVE	AAVE	-7.046	2.341	1957.419	69.821	2124.681	60.877	-0.122	0.077
OKB	OKB	-7.036	2.016	1779.064	139.411	2125.498	134.681	-0.053	0.338
NEAR Protocol	NEAR	-7.022	2.609	1957.512	98.306	2157.687	94.794	-0.163	0.018
Eth. Classic	ETC	-7.072	2.261	1957.794	184.844	2107.841	110.629	-0.121	0.012
Tether Gold	XAUT	-7.104	0.292	1376.323	193.904	1927.243	96.601	-0.044	0.494
PAX Gold	PAXG	-7.112	0.294	1595.150	77.064	1961.342	68.240	-0.026	0.697
KuCoin Token	KCS	-7.170	2.118	1499.946	152.016	1957.102	113.743	-0.106	0.047
Algorand	ALGO	-7.225	2.074	1885.093	88.842	2081.049	130.101	-0.167	0.006
Filcoin	FIL	-7.299	2.213	1989.022	72.529	2166.849	61.891	-0.135	0.051
Wrapped Bitcoin	WBTC	-6.989	1.268	1714.070	274.565	0.000	0.000	-0.159	0.006
Portfolio		-7.049	1.536	1753.464	255.591	2043.905	149.042	-0.111	0.010
Market factor	cmkt							-0.109	0.011
Size factor	csize							0.038	0.382
Momentum factor	cmom							0.004	0.925

This table presents the summary statistics of cryptocurrency variables under study over the sample period from June 2, 2014 to October 28, 2024. The table reports the mean and standard deviation of weekly excess return $R_{i,t}$, dollar trading volume $V_{i,t}$ (in billions), and market capitalization $MV_{i,t}$ (in trillions) for 30 major cryptocurrencies. The row “portfolio” reports aggregate summary statistics for a cryptocurrency portfolio, where we assign the same weight to every cryptocurrency in our sample. The last two columns of the table report the pairwise Pearson cross-correlation coefficients (along with p-values) between the excess returns on individual cryptocurrencies (as well as cryptocurrency risk factors) and the FX volatility risk factor.

Table 2: FX Volatility Risk and the Cross-Section of Cryptocurrency Returns

	FXVOL	CMKT	CSIZE	CMOM	HJ	RMSE	R ² (%)
Panel A: Return Portfolios							
$\hat{\lambda}$	0.044***	-1.333***	0.065***	0.026***	0.031	0.0030	96.4
s.e.	0.004	0.021	0.006	0.002	0.043		
Panel B: Volume Portfolios							
$\hat{\lambda}$	0.056***	-1.289***	-0.047***	-0.174***	0.618	0.0047	95.2
s.e.	0.002	0.015	0.002	0.007	0.043		
Panel C: Size Portfolios							
$\hat{\lambda}$	0.007***	-1.590***	0.004***	0.125***	0.168	0.0029	97.5
s.e.	0.002	0.016	0.001	0.006	0.043		
Panel D: Realized Portfolios							
$\hat{\lambda}$	0.0563***	-1.659***	0.195***	0.544***	1.22	0.005	94.8
s.e.	0.002	0.017	0.013	0.027			
Panel E: Liquidity Portfolios							
$\hat{\lambda}$	0.039***	-1.470***	-0.073***	-0.045***	1.968	0.008	91.4
s.e.	0.001	0.016	0.005	0.007	0.043		

The table presents the price of risk estimates (lambdas, λ , in Equation 12) along with their associated Newey–West standard errors from the second-stage of the FMB asset-pricing tests for a linear four-factor model based on the FX volatility risk factor, while controlling for the market factor, the size factor, and the momentum factor in Liu et al. (2022). The table also reports cross-sectional statistics – the HJ distance measure, RMSE, and R-squared. The test assets are the average equally weighted excess returns of quintile cryptocurrency portfolios sorted based on the prior week excess return (Panel A), trading volume (Panel B), market capitalization (Panel C), volatility (Panel D), and liquidity (Panel E) for the 30 cryptocurrencies listed in Table 1. In each panel, all cryptocurrencies are sorted into five portfolios based on their sorting variable, so that one-fifth of cryptocurrencies with the lowest (highest) values of the sorting variable are allocated to the first (fifth) portfolio. *, **, *** denote statistical significance at 10%, 5% and 1% levels, respectively.

Table 3: FX Volatility Risk: Value-Weighted Evidence

	FXVOL	CMKT	CSIZE	CMOM	HJ	RMSE	R ² (%)
Panel A: Return Portfolios							
$\hat{\lambda}$	0.013***	-1.007***	-0.030***	0.001***	7.736	0.004	98.58
s.e.	0.001	0.001	0.000	0.000	0.043		
Panel B: Volume Portfolios							
$\hat{\lambda}$	0.276***	-0.974***	-0.010***	-0.079***	6.215	0.003	98.99
s.e.	0.002	0.001	0.000	0.001	0.043		
Panel C: Size Portfolios							
$\hat{\lambda}$	0.021***	-1.060***	-0.006***	0.132***	3.552	0.002	99.72
s.e.	0.001	0.001	0.000	0.001	0.043		
Panel D: Volatility Portfolios							
$\hat{\lambda}$	0.070***	-0.976***	-0.078***	-0.111***	1.461	0.001	99.92
s.e.	0.002	0.001	0.001	0.002	0.043		
Panel E: Liquidity Portfolios							
$\hat{\lambda}$	0.119***	-1.019***	-0.058***	-0.032***	0.827	0.001	99.96
s.e.	0.001	0.001	0.000	0.001	0.043		

The table reports the price of risk estimates (along with their associated Newey–West standard errors) and the cross-sectional statistics (the HJ distance measure, RMSE, and R-squared) from the second-stage of the FMB asset-pricing tests for our baseline model based on the FXVOL, CMKT, CSIZE, and CMOM factors. The test assets are the average value weighted excess returns of quintile cryptocurrency portfolios sorted based on the prior week excess return (Panel A), trading volume (Panel B), market capitalization (Panel C), volatility (Panel D), and liquidity (Panel E). In each panel, all sample cryptocurrencies are sorted into five portfolios based on the sorting variable, with the lowest (highest) quintile assigned to the first (fifth) portfolio. *, **, *** denote statistical significance at 10%, 5% and 1% levels, respectively.

Table 4: FX Volatility Risk: Developed and Emerging Markets

	FXVOL _{all}	CMKT	CSIZE	CMOM	HJ	RMSE	R ² (%)
Panel A: Return portfolios							
$\hat{\lambda}$	0.032***	-1.406***	0.032***	0.019***	0.333	0.004	95.92
s.e.	0.004	0.020	0.004	0.002	0.043		
Panel B: Volume portfolios							
$\hat{\lambda}$	0.092***	-1.031***	-0.095***	-0.420***	0.630	0.005	95.14
s.e.	0.003	0.019	0.003	0.013	0.043		
Panel C: Size portfolios							
$\hat{\lambda}$	0.004***	-1.599***	0.005***	0.128***	0.187	0.003	97.58
s.e.	0.001	0.015	0.001	0.005	0.043		
Panel D: Realized volatility portfolios							
$\hat{\lambda}$	0.057***	-1.658***	0.238***	0.604***	0.512	0.003	97.39
s.e.	0.002	0.017	0.012	0.026	0.043		
Panel E: Liquidity portfolios							
$\hat{\lambda}$	0.028***	-1.510***	-0.063***	-0.018**	2.068	0.008	90.39
s.e.	0.001	0.016	0.005	0.007	0.043		

The table presents the asset pricing results obtained from the second-pass FMB regression for our baseline model based on the $FXVOL_{all}$ (calculated using data from both developed and emerging market currencies), CMKT, CSIZE, and CMOM factors. The test assets are the average equally weighted excess returns of quintile cryptocurrency portfolios sorted based on the prior week excess return (Panel A), trading volume (Panel B), market capitalization (Panel C), volatility (Panel D), and liquidity (Panel E). Each panel reports the estimated risk prices, the Newey–West standard errors, and the cross-sectional statistics (the HJ distance measure, RMSE, and R-squared). *, **, *** denote statistical significance at 10%, 5% and 1% levels, respectively.

Table 5: FX Volatility Risk: Subsample Analysis

	FXVOL	CMKT	CSIZE	CMOM	HJ	RMSE	R ² (%)
Panel A: Pre-COVID (June 2, 2014 - January 31, 2020)							
$\hat{\lambda}$	0.009***	-0.942***	0.030***	0.014***	0.120	0.004	93.25
s.e.	0.001	0.014	0.005	0.002	0.058		
Panel B: Post-COVID (February 1, 2020 - October 31, 2024)							
$\hat{\lambda}$	0.014***	-1.878***	-0.005**	0.004***	1.221	0.003	97.85
s.e.	0.002	0.028	0.002	0.001	0.063		
Panel C: Bull Market (January 1, 2020 - November 14, 2021)							
$\hat{\lambda}$	0.042***	-1.095***	0.017***	0.013***	1.046	0.003	97.56
s.e.	0.003	0.064	0.004	0.002	0.101		
Panel D: Bear Market (November 15, 2021 - October 31, 2024)							
$\hat{\lambda}$	0.005***	-2.122***	-0.008***	0.002**	0.707	0.002	99.51
s.e.	0.002	0.027	0.002	0.001	0.080		

The table presents the lambda estimates (along with Newey–West standard errors) from the second-stage of the FMB asset-pricing tests for a linear four-factor model (based on the FXVOL, CMKT, CSIZE, and CMOM factors) across four subsample periods: (1) Pre-COVID period (June 2, 2014 - January 31, 2020), (2) Post-COVID period (February 1, 2020 - October 31, 2024), (3) Bull market period (January 1, 2020 - November 14, 2021), and (4) Bear market period (November 15, 2021 - October 31, 2024). The cross-sectional statistics (the HJ distance measure, RMSE, and R-squared) are reported in each panel. The test assets are the average equally weighted excess returns of quintile portfolios sorted based on the prior week cryptocurrency excess return. *, **, *** denote statistical significance at 10%, 5% and 1% levels, respectively.

Table 6: FX Volatility Risk: Crypto-Level Analysis

	FXVOL	CMKT	CSIZE	CMOM	HJ	RMSE	R ² (%)
Panel A: 30 cryptocurrencies (Whole sample)							
$\hat{\lambda}$	0.009***	-1.576***	0.014**	0.057***	25.306	0.019	88.8
s.e.	0.002	0.017	0.006	0.010	0.069		
Panel B: 20 cryptocurrencies							
$\hat{\lambda}$	0.007***	-1.560***	0.018***	0.055***	7.697	0.016	91.97
s.e.	0.002	0.017	0.006	0.010	0.068		
Panel C: 10 cryptocurrencies							
$\hat{\lambda}$	0.113***	-1.400***	0.189***	0.403***	6.861	0.009	96.45
s.e.	0.015	0.041	0.022	0.043	0.066		
Panel D: 5 cryptocurrencies							
$\hat{\lambda}$	0.022***	-1.656***	-0.046***	0.198***	2.958	0.002	99.69
s.e.	0.004	0.030	0.010	0.011	0.066		

The table presents the crypto-level asset pricing results for our baseline model based on the FXVOL, CMKT, CSIZE, and CMOM factors. The test assets consist of the 30 (Panel A), 20 (Panel B), 10 (Panel C), and 5 (Panel D) largest cryptocurrencies by market capitalization. Each panel reports the estimated risk prices (along with Newey–West standard errors) in addition to the cross-sectional statistics (the HJ distance measure, RMSE, and R-squared). *, **, *** denote statistical significance at 10%, 5% and 1% levels, respectively.

Table 7: FX Volatility Risk: Controlling for Crypto-Specific Risks

	stat.	FXVOL	CMKT	CSIZE	CMOM	CMKTDR	CVOL	CLIQ	UCRY	GPR
1	$\hat{\lambda}$		-2.162***	-0.006**	0.002*	0.613***				
	s.e.		0.058	0.002	0.001	0.060				
2	$\hat{\lambda}$	0.015**	-2.036***	0.017	0.010**	0.525***				
	s.e.	0.007	0.109	0.012	0.004	0.098				
3	$\hat{\lambda}$		-1.696***	-0.040***	-0.004***		23.828***			
	s.e.		0.023	0.005	0.001		2.118			
4	$\hat{\lambda}$	0.050***	-1.277***	0.080***	0.030***		-4.518			
	s.e.	0.010	0.084	0.024	0.007		5.920			
5	$\hat{\lambda}$		-1.604***	-0.016***	-0.002*			0.0114***		
	s.e.		0.018	0.003	0.001			0.001		
6	$\hat{\lambda}$	0.051***	-1.289***	0.078***	0.030***			0.000		
	s.e.	0.010	0.068	0.020	0.007			0.003		
7	$\hat{\lambda}$		-1.465***	-0.013***	0.008***				-0.289***	
	s.e.		0.015	0.003	0.001				0.025	
8	$\hat{\lambda}$	0.055***	-1.302***	0.084***	0.030***				0.084	
	s.e.	0.016	0.051	0.029	0.007				0.112	
9	$\hat{\lambda}$		-1.498***	0.001	0.003***					-0.511***
	s.e.		0.015	0.002	0.001					0.184
10	$\hat{\lambda}$	0.044***	-1.333***	0.064***	0.025***					0.152
	s.e.	0.004	0.021	0.005	0.002					0.187

The table reports the second-stage results of the FMB asset pricing test (λ estimates, Newey–West standard errors, and the cross-sectional statistics) for an augmented version of our four-factor model (based on FXVOL, CMKT, CSIZE and CMOM factors), wherein we separately control for five cryptocurrency-specific risk factors: (1) the cryptocurrency downside market risk factor (CMKTDR) from [Zhang et al. \(2021\)](#) and [Dobrynskaya \(2024\)](#); (2) the cryptocurrency volatility factor in [Lee and Wang \(2024\)](#); (3) the cryptocurrency liquidity factor in [Brauneis et al. \(2021\)](#); (4) the cryptocurrency uncertainty index in [Lucey et al. \(2022\)](#); and (5) the geopolitical risk index (GPR) in [Caldara and Iacoviello \(2022\)](#). The test assets are the average equally weighted excess returns of quintile cryptocurrency portfolios sorted on the prior week cryptocurrency excess return. *, **, *** denote statistical significance at 10%, 5% and 1% levels, respectively.

Table 8: FX Volatility Risk: Controlling for Equity and Bond-Specific Risks

stat.	FXVOL	CMKT	CSIZE	CMOM	EQVOL	BOVOL	WOEQ	WOBO	USEQ	USBO	VIX
1 $\hat{\lambda}$	0.041***	-1.341***	0.059***	0.024***	0.054***						
s.e.	0.006	0.022	0.010	0.003	0.006						
2 $\hat{\lambda}$	0.044***	-1.330***	0.066***	0.026***		0.012***					
s.e.	0.004	0.021	0.006	0.002		0.005					
3 $\hat{\lambda}$	0.042***	-1.342***	0.064***	0.025***			-0.020***				
s.e.	0.004	0.023	0.006	0.002			0.006				
4 $\hat{\lambda}$	0.033**	-1.386***	0.057***	0.019*				0.006			
s.e.	0.016	0.077	0.013	0.010				0.008			
5 $\hat{\lambda}$	0.044***	-1.340***	0.066***	0.025***					-0.020***		
s.e.	0.004	0.022	0.006	0.002					0.007		
6 $\hat{\lambda}$	0.045***	-1.348***	0.062***	0.028***						-0.002	
s.e.	0.004	0.028	0.007	0.004						0.005	
7 $\hat{\lambda}$	0.045***	-1.329***	0.064***	0.026***							-0.029
s.e.	0.004	0.022	0.006	0.002							0.050

The table reports the second-stage results of the FMB asset pricing test (λ estimates, Newey–West standard errors, and the cross-sectional statistics) for an augmented version of our four-factor model (based on FXVOL, CMKT, CSIZE and CMOM factors), wherein we separately control for seven equity- and bond-specific variables: (1) equity market volatility (EQVOL); (2) bond market volatility (BOVOL); (3) the MSCI World stock market index return (WOEQ); (4) the Citibank World Government Bond Index return (WOBO); (5) the US Datastream equity index return (USEQ); (6) the US 5-year Datastream government bond index return (USBO); and (7) the change in the VIX index. Both the EQVOL and BOVOL factors are constructed using our sample of 10 developed countries: Australia, Canada, Germany, Japan, New Zealand, Norway, Switzerland, Sweden, Denmark, and the United Kingdom. The test assets are the average equally weighted excess returns of quintile cryptocurrency portfolios sorted on the prior week cryptocurrency excess return. *, **, *** denote statistical significance at 10%, 5% and 1% levels, respectively.

Table 9: Cryptocurrency and Currency Asset Pricing under Shared Risks

Panel A: Crypto portfolios									
statistic	CMKT	CSIZE	CMOM	DOL	CAR	MOM	FXVOL	CVOL	CLIQ
1 $\hat{\lambda}$	-1.509***	0.002	0.004***	-0.002					
s.e.	0.016	0.002	0.001	0.002					
2 $\hat{\lambda}$	-1.495***	0.010***	0.002*		-0.022***				
s.e.	0.016	0.003	0.001		0.005				
3 $\hat{\lambda}$	-1.485***	-0.006**	0.001			-0.030***			
s.e.	0.015	0.002	0.001			0.003			
4 $\hat{\lambda}$	-1.333***	0.065***	0.026***				0.044***		
s.e.	0.006	0.002	0.021				0.004		
5 $\hat{\lambda}$	-1.696***	-0.040***	-0.004***					23.828***	
s.e.	0.023	0.005	0.001					2.118	
6 $\hat{\lambda}$	-1.604***	-0.016***	-0.002*						0.0114***
s.e.	0.018	0.003	0.001						0.001
Panel B: Currency portfolios									
statistic	DOL	CAR	MOM	CMKT	CSIZE	CMOM	FXVOL	CVOL	CLIQ
1 $\hat{\lambda}$	-0.001***	0.001*	0.001***	-1.5105***					
s.e.	0.000	0.001	0.000	0.256					
2 $\hat{\lambda}$	-0.0009***	0.0035***	0.0012***		0.0483***				
s.e.	0.000	0.001	0.000		0.014				
3 $\hat{\lambda}$	-0.0009***	0.0050***	0.0012***			0.0823***			
s.e.	0.000	0.001	0.000			0.016			
4 $\hat{\lambda}$	-0.001***	0.003***	0.001***				-0.008***		
s.e.	0.000	0.001	0.000				0.001		
5 $\hat{\lambda}$	-0.001***	0.003***	0.001***					3.293***	
s.e.	0.0001	0.001	0.000					0.720	
6 $\hat{\lambda}$	-0.0009***	0.0024***	0.0012***						0.0012***
s.e.	0.000	0.001	0.000						0.000

The Table compares the asset pricing results across cryptocurrency returns (Panel A) and currency returns (Panel B) as test assets. Panel A reports results for six different specifications of a four-factor model, which differ in the choice of risk factors from the set DOL, CAR, MOM, FXVOL, CVOL, and CLIQ, while controlling for CMKT, CISIZE, and CMOM factors in all model specifications. Panel B presents results for five different specifications, where each specification varies the risk factor set among CMKT, CSIZE, CMOM, FXVOL, CVOL, and CLIQ, while controlling for DOL, CAR, and MOM factors in all model specifications. In each panel, we report the price of risk estimates along with Newey–West standard errors from the second-stage of the FMB asset-pricing tests. *, **, *** denote statistical significance at 10%, 5% and 1% levels, respectively.

Table 10: The Cross-Asset Spillover Effects of FX Volatility: Time-Series Evidence

Panel A: Cryptocurrency Returns						
statistic	FXVOL _t	CMKT _t	CSIZE _t	CMOM _t	$r_{i,t-1}^{crypto}$	$r_{j,t-1}^{FX}$
1 coeff.	-4.432***	0.043***	0.350***	0.324***		
s.e.	0.339	0.002	0.026	0.025		
2 coeff.	-3.508***				0.765***	0.231
s.e.	0.441				0.030	0.319
3 coeff.	-2.275***	0.043***	0.194***	0.185***	0.439***	-0.286
s.e.	0.301	0.000	0.023	0.022	0.026	0.221
Panel B: Currency Returns						
statistic	FXVOL _t	CMKT _t	CSIZE _t	CMOM _t	$r_{i,t-1}^{crypto}$	$r_{j,t-1}^{FX}$
1 coeff.	-0.009	0.001***	0.011***	-0.000		
s.e.	0.053	0.000	0.004	0.004		
2 coeff.	0.126**				0.020***	-0.008
s.e.	0.055				0.004	0.048
3 coeff.	0.069	0.001***	0.005	-0.005	0.016***	0.002
s.e.	0.058	0.000	0.004	0.004	0.005	0.042

The table presents the time-series coefficient estimates (and the associated standard errors) of the M, A, and R-matrices from the mean equation of the VAR GARCH-M model in Equation 13. The M-matrix captures the joint spillover of FX volatility to cryptocurrency returns and currency returns; the A-matrix allows the two market returns to be simultaneously influenced by cryptocurrency risk factors (CMKT, CSIZE, and CMOM); and the R-matrix controls for cross-market return spillovers from one market to the other, as well as responses to each market's own lagged returns. The test assets are cryptocurrency (Panel A) and currency (Panel B) portfolio returns, each assigning equal weights to all cryptocurrencies and developed-market currencies, respectively. *, **, *** denote statistical significance at 10%, 5% and 1% levels, respectively.

Table 11: Cross-Asset Time-Series Momentum

Panel A: Top five cryptocurrencies				
	MOM(10)	MOM(20)	MOM(50)	MOM(100)
Bitcoin				
Mean	7.01***	7.00***	6.97***	6.97***
St. Dev.	0.01	0.01	0.01	0.01
Skewness	0.15	0.15	0.14	0.14
Ethereum				
Mean	6.85***	6.85***	6.85***	6.88***
St. Dev.	0.02	0.02	0.02	0.02
Skewness	-0.55	-0.56	-0.55	-0.36
XRP				
Mean	6.89***	6.91***	6.88***	6.84***
St. Dev.	0.02	0.02	0.02	0.03
Skewness	-1.43	-1.33	-1.37	-1.34
Binance				
Mean	6.97***	6.96***	6.96***	6.96***
St. Dev.	0.02	0.02	0.02	0.02
Skewness	0.15	0.14	0.14	0.14
Solana				
Mean	6.66***	6.65***	6.66***	6.69***
St. Dev.	0.03	0.03	0.03	0.03
Skewness	-0.46	-0.47	-0.46	-0.44
Panel B: Equally-Weighted Portfolio (XTSMOM)				
	MOM(10)	MOM(20)	MOM(50)	MOM(100)
Mean	6.96***	6.96***	6.94***	6.95***
St. Dev.	0.02	0.02	0.02	0.02
Skewness	0.07	0.06	0.07	0.18

The table shows the results for the cross-asset time-series momentum strategy where investors long (short) cryptocurrencies conditional on a trading signal which equals +1 (-1) if the past weekly cryptocurrency excess return is positive (negative) and that of the FX volatility is negative (positive). Otherwise, traders take no position. The return on this strategy $r_{i,t+1}^{TSMOM}$ equals the actual cryptocurrency excess return multiplied by the trading signal (see Equation 14). The table reports three performance measures (mean, volatility, and skewness) for two tests assets – the five largest individual cryptocurrencies by market capitalization (Panel A) and an equally weighted cryptocurrency portfolio (Panel B). These statistics are performed on FX volatilities for lags 10, 20, 50, and 100. *, **, *** denote statistical significance 10%, 5% and 1% levels, respectively.

Table 12: Cross-Asset Time-Series Moving-Average

Panel A: Top five cryptocurrencies				
	MA(10,50)	MA(10,100)	MA(20,50)	MA(20,100)
Bitcoin				
Mean	4.72***	5.03***	4.79***	4.94***
St. Dev.	0.03	0.03	0.03	0.03
Skewness	-0.44	-0.56	-0.46	-0.52
Ethereum				
Mean	4.88***	4.97***	4.93***	4.90***
St. Dev.	0.03	0.04	0.04	0.04
Skewness	-0.33	-0.35	-0.34	-0.31
XRP				
Mean	4.61***	4.90***	4.67***	4.81***
St. Dev.	0.04	0.04	0.04	0.04
Skewness	-0.19	-0.34	-0.22	-0.29
Binance				
Mean	4.57***	3.85***	4.55***	3.66***
St. Dev.	0.04	0.04	0.04	0.04
Skewness	-0.26	0.15	-0.23	0.25
Solana				
Mean	4.01***	3.92***	4.11***	3.86***
St. Dev.	0.04	0.04	0.04	0.04
Skewness	0.05	0.04	0.01	0.07
Panel B: Equally-Weighted Portfolio (XTSMOM)				
	MA(10,50)	MA(10,100)	MA(20,50)	MA(20,100)
Mean	4.68***	4.96***	4.74***	4.86***
St. Dev.	0.03	0.03	0.03	0.03
Skewness	-0.38	-0.46	-0.40	-0.41

The table shows the results for the cross-asset time-series moving-average strategy where investors long (short) cryptocurrencies conditional on a trading signal which equals +1 (−1) if the past weekly cryptocurrency excess return is positive (negative) and the MA-based trend following indicator, as defined in Equation 15, equals +1 (−1). Otherwise, traders take no position. The return on this trading strategy is defined in Equation (16). The table reports three performance measures (mean, volatility, and skewness) for two tests assets – the five largest individual cryptocurrencies by market capitalization (Panel A) and an equally weighted cryptocurrency portfolio (Panel B). The MA results are reported with short MA = 10 and 20, and long MA = 50 and 100. *, **, *** denote statistical significance 10%, 5% and 1% levels, respectively.